Problem 2. (**QR decomposition for ill-conditioned matrices**)(25 points)

Let $\boldsymbol{e} = \frac{1}{\sqrt{n}} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^n$, $\boldsymbol{\alpha} = \boldsymbol{e}\sqrt{1-\varepsilon}$ ($\varepsilon = 10^{-6}$) and $\boldsymbol{A} = I_n - \boldsymbol{\alpha}\boldsymbol{\alpha}^T$. As an ill-conditioned matrix with a condition number growing exponentially with the dimension n, it serves as an ideal test case to verify the stability of numerical methods. This experiment focuses on four QR decomposition methods: Classical Gram-Schmidt (CGS), Modified Gram-Schmidt (MGS), Householder Reflections, and Givens Rotations. For $n = 2, 3, \ldots, 20$, we aim to explore the decomposition stability of these methods and generate two plots as specified below:

(1) the relationship between the decomposition accuracy $\|\mathbf{QR} - \mathbf{A}\|_2$ of the four methods and the dimension n;

(2) the relationship between the orthogonality error $\|\mathbf{Q^T Q} - \mathbf{I}_n\|_2$ of the four methods and the dimension n.